DEPARTMENT OF MATHEMATICS
UNIVERSITY COLLEGE OF ENGINEERING (A) – O.U.,HYD - 7
B.E. (**AI&ML**) IV- SEMESTER, CLASS TEST- I
Engineering Mathematics – IV
(Numerical Methods)

1 00523729008

TIME: 1hr 30 min **Max. Marks: 30**

	PART – A (Answer **all questions**) $5\times2=10$	Marks	CO	BT
1 a)	obtain a partial differential equation that governs the family of surfaces $z=(x-\alpha)^2+(y-\beta)^2$	2	1	L2
b)	Find particular integral of $(D^2+DD'-6D'^2)z=\cos(3x+y)$	2	1	L2
c)	Solve $(D^2-DD'-6D'^2)z=0$	2	1	L2
d)	Classify the partial differential equation $\frac{\partial^2 u}{\partial x^2}+3\frac{\partial^2 u}{\partial y^2}-\frac{\partial u}{\partial x}=0$	2	2	L1
e)	Use D'Alemberts solution to find the solution of the IVP defining the vibrations of an infinitely long elastic string when $f(x)=\sin 2x,\quad g(x)=\cos 2x$.	2	2	L3
	PART - B (Answer any **TWO** questions) $2\times10=20$			
2	**a.** solve the differential equation $2yzp+zxq=3xy$	5	1	L3
	b. solve $\frac{\partial^3 z}{\partial x^3}-3\frac{\partial^3 z}{\partial x^2\partial y}+4\frac{\partial^3 z}{\partial y^3}=e^{x+2y}$	5	1	L3
3	**a.** solve the differential equation $3\frac{\partial u}{\partial x}+2\frac{\partial u}{\partial y}=0$, *where* $u(x,0)=4e^{-x}$ by the method of separation of variables.	4	2	L3
	b. A tightly stretched string with fixed end points $x=0$ *and* $x=l$ is initially in a position given by $u(x,0)=u_0\sin^3(\frac{\pi x}{l})$. If it is released from rest from this position, find displacement u at any distance x from one end and at any time t	6	2	L4
4	**a.** Solve $\frac{\partial^2 z}{\partial x^2}+3\frac{\partial^2 z}{\partial x\partial y}+2\frac{\partial^2 z}{\partial y^2}=x+y$	5	1	L3
	b. Find the solution of 1-D wave equation in all cases.	5	2	L4

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